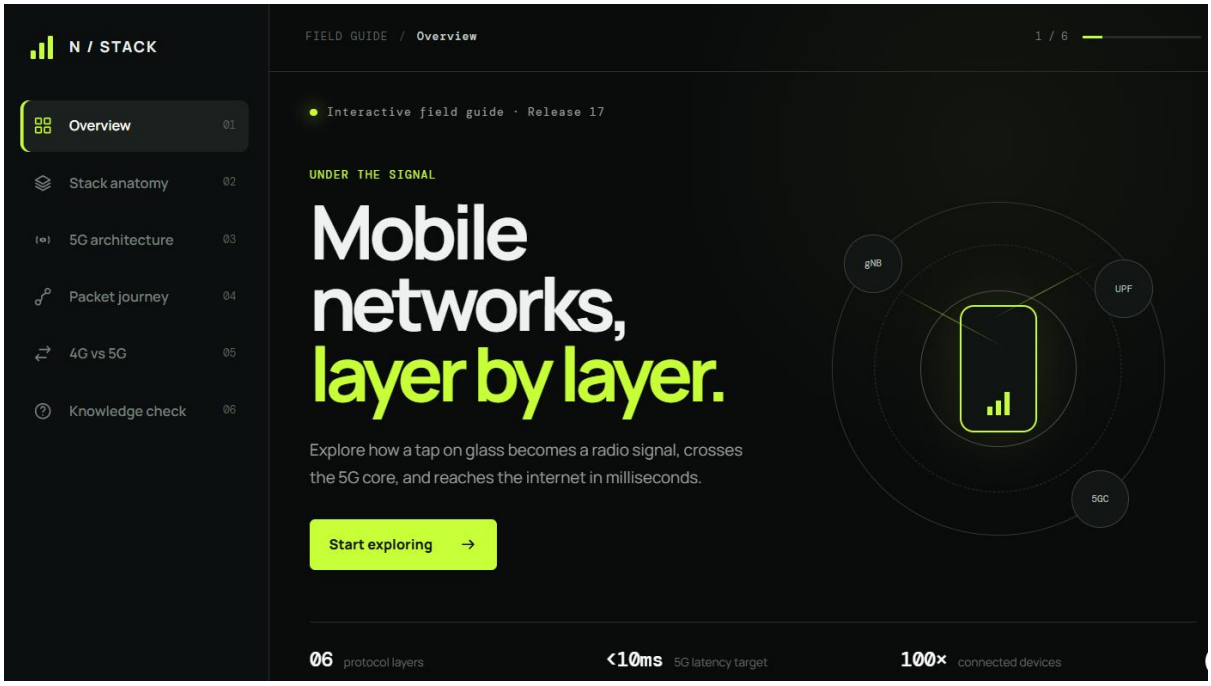
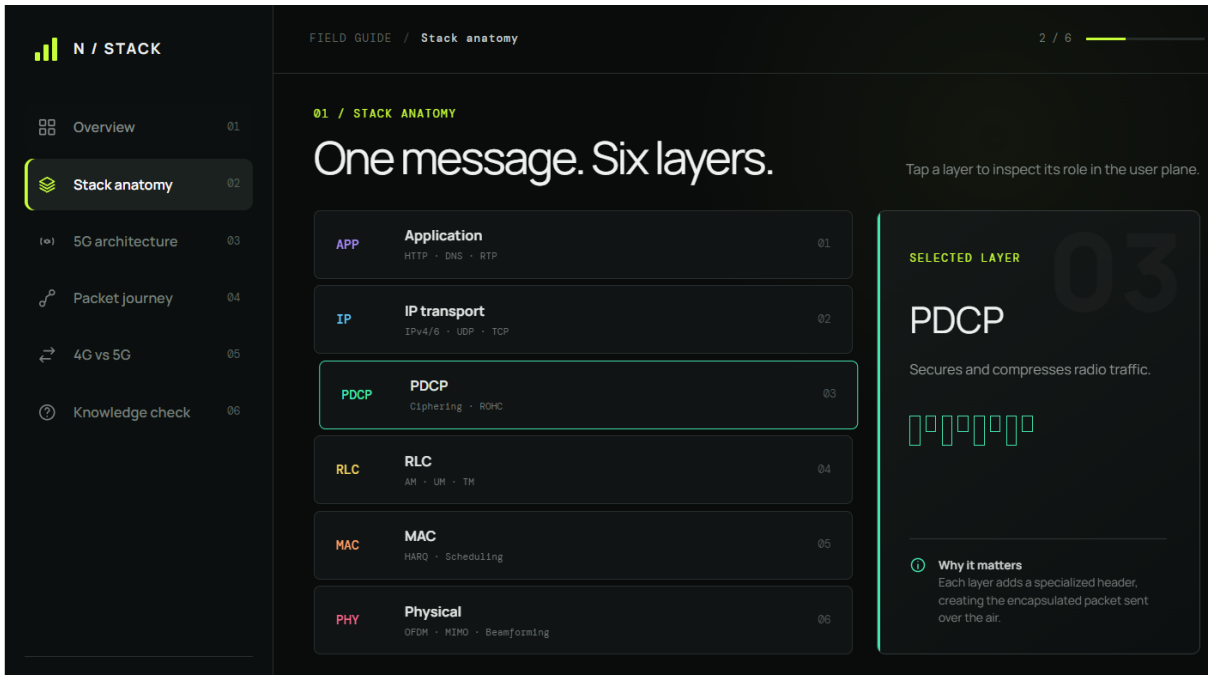
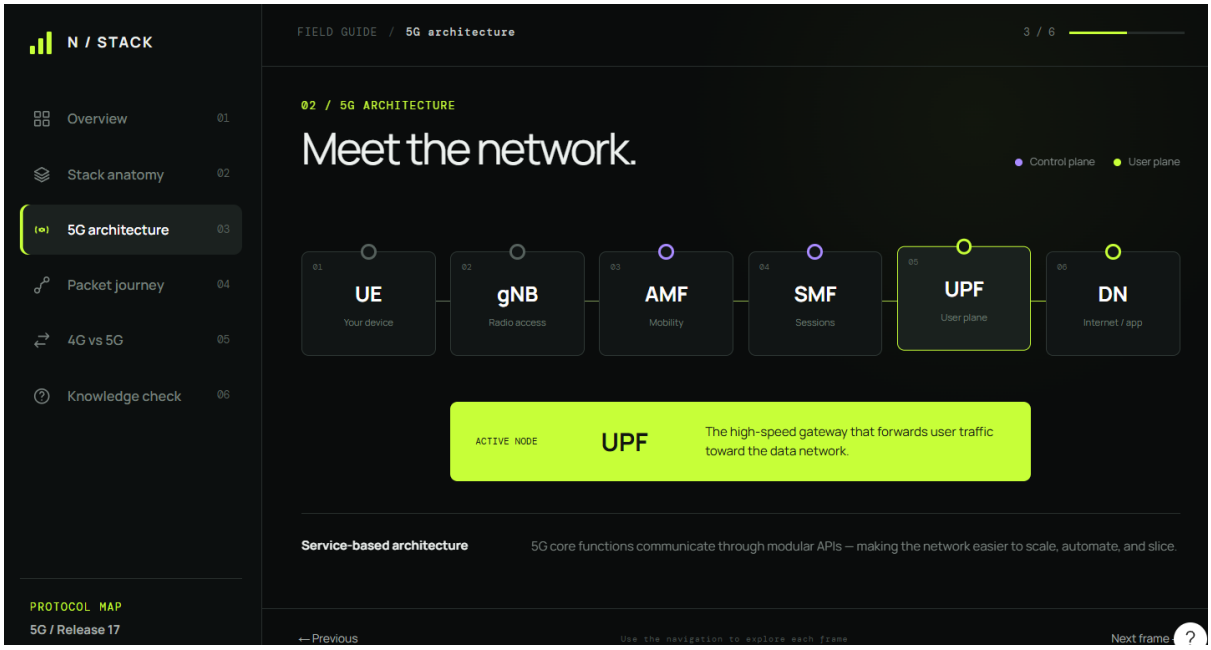
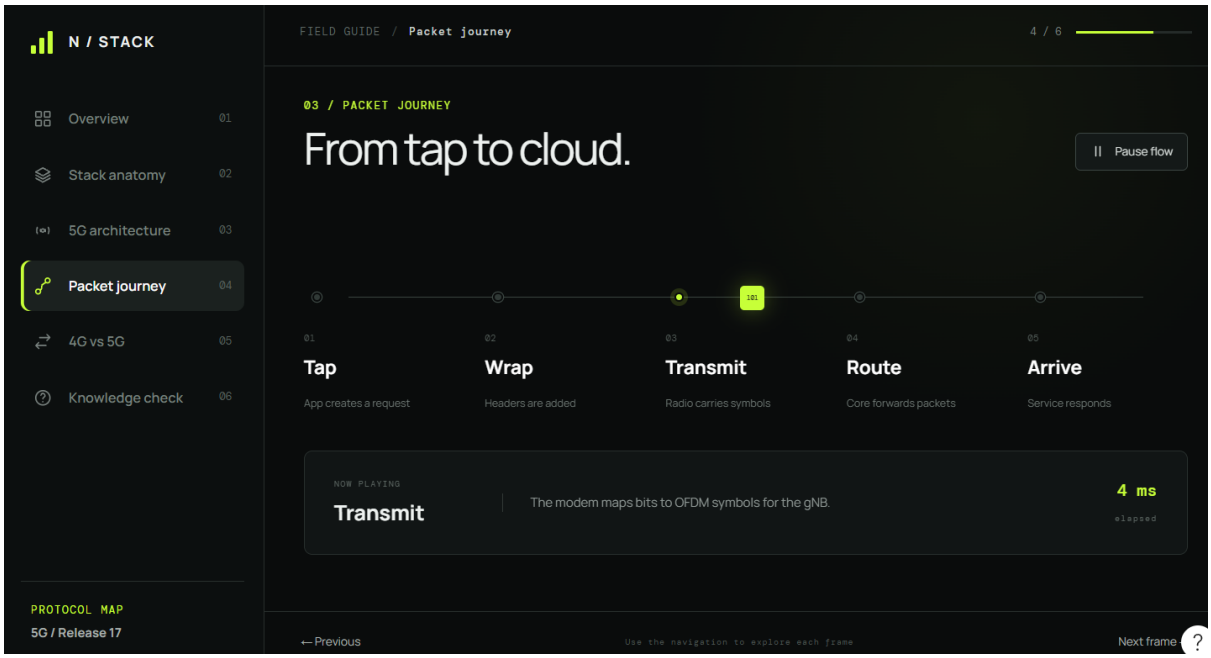
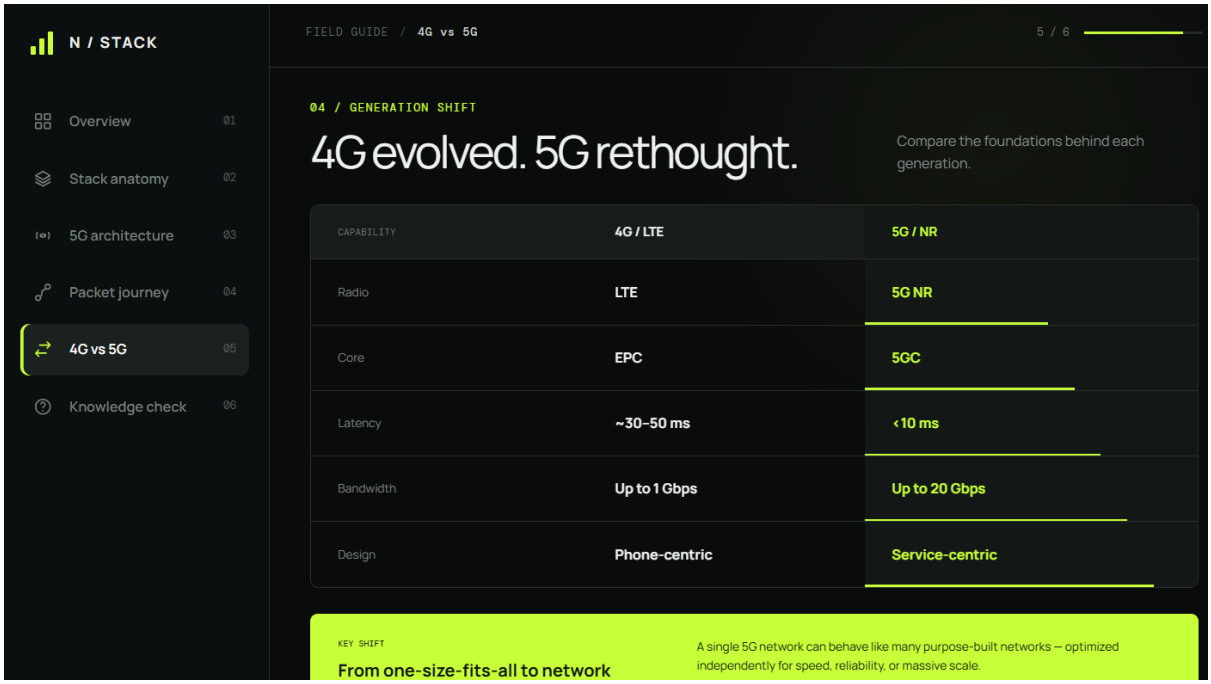
Experiment 4

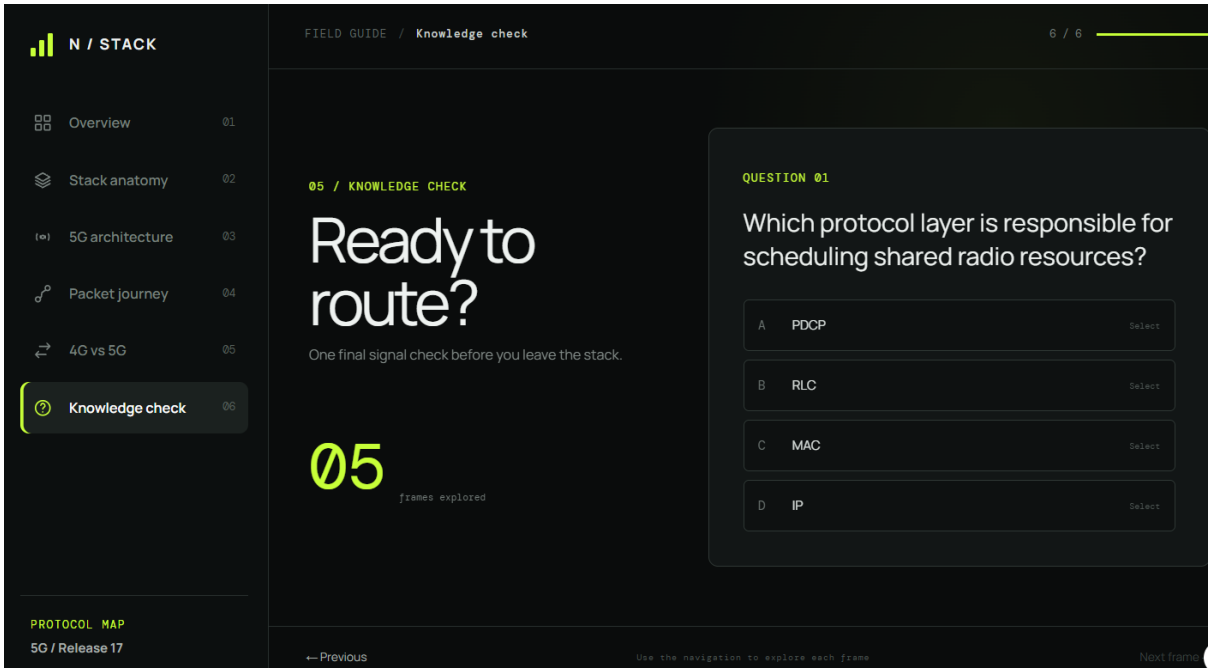





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5G architecture 03
Packet journey 04
4G vs 5G 05
Knowledge check 06
PROTOCOL MAP
5G / Release 17
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02 / 5G ARCHITECTURE
Meet the network.
Control plane
User plane
01 UE Your device
02 gNB Radio access
03 AMF Mobility
04 SMF Sessions
05 UPF User plane
06 DN Internet / app
ACTIVE NODE
UPF
The high-speed gateway that forwards user traffic toward the data network.
Service-based architecture
5G core functions communicate through modular APIs — making the network easier to scale, automate, and slice.
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Use the navigation to explore each frame
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03 / PACKET JOURNEY
From tap to cloud.
Pause flow
01 Tap App creates a request
02 Wrap Headers are added
03 Transmit Radio carries symbols
04 Route Core forwards packets
05 Arrive Service responds
NOW PLAYING
Transmit
The modem maps bits to OFDM symbols for the gNB.
4 ms
elapsed
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Use the navigation to explore each frame
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04 / GENERATION SHIFT
4G evolved. 5G rethought.
Compare the foundations behind each generation.
CAPABILITY
4G / LTE
5G / NR
Radio
LTE
5G NR
Core
EPC
5GC
Latency
~30–50 ms
<10 ms
Bandwidth
Up to 1 Gbps
Up to 20 Gbps
Design
Phone-centric
Service-centric
KEY SHIFT
From one-size-fits-all to network
A single 5G network can behave like many purpose-built networks — optimized independently for speed, reliability, or massive scale.


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05 / KNOWLEDGE CHECK
Ready to route?
One final signal check before you leave the stack.
05
frames explored
QUESTION 01
Which protocol layer is responsible for scheduling shared radio resources?
A PDCP Select
B RLC Select
C MAC Select
D IP Select
← Previous
Use the navigation to explore each frame
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